

Loneliness as Social Reward Prediction Error

A Developmental Neurocognitive Framework of Adolescent Reward Recalibration

Xueru Zhao

Revised: February 2026

Abstract

I came to this topic partly because I kept running into a gap in the explanations I read. The most common accounts of adolescent loneliness treat it primarily as an emotional experience — a kind of social pain that motivates reconnection. That framing is not wrong. But when I started reading the reinforcement learning literature alongside the loneliness literature, I started wondering whether something more specific was happening mechanistically, something that could actually be measured rather than just described.

In this paper, I propose a preliminary framework suggesting that chronic adolescent loneliness can be modeled, at least in part, as a persistent pattern of negative social reward prediction error (sRPE). The basic idea is that when there is a sustained gap between expected and received social reward, the brain does not only generate distress — it may also generate a learning signal that progressively recalibrates how social outcomes are anticipated. During adolescence, when dopaminergic plasticity appears to be especially pronounced, that recalibration could have effects that are larger and more lasting than at other developmental periods.

I am calling this account the Adolescent Social Reward Recalibration Framework (ASRRF). It is meant as a theoretical proposal, not a completed theory. The main reason I think it is worth developing is that it generates predictions that are, in principle, empirically separable from those of existing accounts: (i) chronic loneliness should preferentially attenuate ventral striatal response to anticipated rather than received social reward; (ii) that attenuation should track reinforcement-learning parameters such as learning rate α rather than trait threat sensitivity; and (iii) developmentally timed manipulations of social expectation should produce larger recalibration effects in mid-adolescence than in adulthood. These are not guaranteed to hold up. But they are testable in ways that more purely affective accounts of loneliness are not.

Introduction

Loneliness is one of those topics that feels straightforward until you sit down and try to explain it precisely. The intuitive story — you feel lonely when you lack social connection — does not really account for the empirical patterns. People can feel deeply lonely in crowded environments, and others can live relatively isolated lives without experiencing loneliness at all. That asymmetry suggests loneliness is not tracking objective social density. It is tracking something internal: some evaluation of whether social experience is matching what was expected.

That realization sent me in a somewhat unexpected direction. If loneliness is fundamentally about unmet expectation, then it is not just an affective state — it is a cognitive one, and its effects should run through the systems the brain uses to encode and update expectations. Which in the reward learning literature means dopaminergic systems and their role in computing

prediction error. I do not want to overstate how directly that mapping works. Social belonging is not the same as receiving a monetary reward in a task, and I will return to that caveat several times in this paper. But the basic logic seems worth pursuing: if expectation violation drives loneliness, and if the brain has a general mechanism for responding to expectation violations in learning-relevant ways, then loneliness and that mechanism might be more connected than current accounts acknowledge.

The developmental context matters because adolescence is not a neutral period for these systems. There is fairly consistent evidence that reward-related neural circuits — particularly dopaminergic pathways and frontostriatal connections — are undergoing significant refinement during adolescence (Forbes & Dahl, 2012; Larsen & Luna, 2018). This is also the period when social feedback becomes especially salient as a source of reinforcement. The combination of heightened reward sensitivity and elevated social stakes creates a situation where repeated social mismatches might leave a deeper trace on motivational systems than they would at other ages. That is the core developmental claim, and I think it is plausible even if the direct evidence is limited.

Departure from the evolutionary account. The account I develop here differs from what is probably the most influential existing framework — the evolutionary loneliness model developed by Cacioppo and colleagues (Cacioppo & Hawkley, 2009; Cacioppo, Cacioppo, & Boomsma, 2014). That account treats loneliness as an evolved aversive signal, analogous to hunger or pain, whose function is to motivate social reconnection and whose primary cognitive signature is hypervigilance to social threat. I find that account compelling in a number of ways. It explains the short-term motivational properties of loneliness reasonably well, and it has been tested against a substantial body of data.

Where I think it is less complete is in explaining what happens when the signal fails to achieve its goal — when loneliness persists despite motivational pressure toward reconnection, and when the system itself starts to change as a result. The ASRRF focuses specifically on that recalibration process. Rather than treating loneliness primarily as an alarm that triggers behavioral correction, I want to model it as a learning signal that updates the system's parameters. Whether that addition is necessary, or whether it is captured within an expanded evolutionary account, is something I genuinely do not know. What I think is true is that the prediction-error framing generates different empirical predictions, and that is reason enough to develop it.

Three differences from Cacioppo's account are worth specifying. Mechanistically, an evolutionary alarm account predicts heightened threat detection and attentional capture by social cues; a prediction-error account predicts changes in dopaminergic learning-rate parameters and altered incentive-salience encoding in ventral striatum and medial prefrontal cortex (Schultz, 2016; Hauser et al., 2015). Developmentally, the evolutionary account is largely lifespan-invariant; the present account is bounded by a plasticity window in adolescence during which these parameters are more susceptible to recalibration (Larsen & Luna, 2018). Predictively, the two accounts diverge on which variables should most strongly covary with chronic loneliness — threat-sensitivity measures versus anticipatory reward signals and formal RL parameters.

The key empirical question is whether chronic loneliness primarily diminishes anticipation of social reward, or primarily amplifies sensitivity to its absence. These are not identical predictions, and they imply different neural signatures and different intervention strategies. I will

argue the first is more diagnostic of the account I am proposing, though I acknowledge the evidence does not yet cleanly resolve this.

Developmental neuroscience provides useful context here. Neuroimaging studies consistently show heightened ventral striatal and medial prefrontal responsivity to social feedback and peer evaluation during adolescence (Galván, 2010; Guyer, Silk, & Nelson, 2016; Schreuders et al., 2018). As social interaction becomes a primary source of reinforcement during this period, acceptance and rejection acquire unusual motivational weight. Disruptions in perceived social reward may therefore disproportionately affect the motivational systems still being calibrated.

Central to the framework is reward prediction error (RPE) — a signal encoding the discrepancy between expected and received outcomes. RPE signals, primarily mediated by dopaminergic pathways, guide updating of value representations and shape future motivation (Hauser et al., 2015). During adolescence, ongoing changes in dopaminergic function and prefrontal-striatal connectivity may amplify sensitivity to these signals, making the cumulative effects of repeated social mismatch more consequential than they would be in adulthood.

The sections that follow develop each piece of this argument in more detail. Section 1 reviews loneliness as a neurocognitive construct. Section 2 covers adolescent reward system development. Section 3 addresses prediction error and adaptive learning. Section 4 presents the ASRRF as a preliminary framework. Sections 5 and 6 discuss behavioral implications and future directions, including limitations I think are important to be explicit about.

1 Loneliness as a Neurocognitive Construct

Loneliness has traditionally been framed as an affective response to social disconnection. That framing is not inaccurate — loneliness is certainly affective. But I think treating it primarily as an emotion tends to obscure the cognitive machinery that produces it. If loneliness were purely emotional, we might expect its occurrence to correlate straightforwardly with objective social isolation. It does not. That discrepancy suggests that loneliness is generated not by circumstances per se, but by the evaluation of those circumstances relative to some internal standard.

That evaluative character means loneliness is tied to systems responsible for valuation, prediction, and adaptive learning — not just to affective processing. The following subsections try to spell out what that implies.

1.1 Subjective Appraisal and Social Expectation

Contemporary accounts of loneliness have increasingly emphasized that subjective appraisal, rather than objective social circumstance, is the proximal cause of the experience. People compare their perceived social environment against internal standards of belonging, acceptance, and interpersonal value. Loneliness arises when that comparison produces a significant discrepancy. What counts as a significant discrepancy will vary across individuals and contexts, which is part of why two people in similar situations can have such different experiences.

This expectation-based view aligns loneliness with predictive processing accounts of cognition more broadly, in which the brain continuously evaluates incoming information against prior beliefs. When expectations are violated, a signal is generated. The interesting question — for this paper, at least — is what kind of signal loneliness generates, and what the brain does with it.

I should note that the expectation-violation framing is not unique to the account I am proposing. It is also present in Cacioppo's evolutionary model and in cognitive accounts of loneliness more generally (Cacioppo & Hawkley, 2009). What the ASRRF adds, or tries to add, is a specific claim about the downstream learning consequences of that signal — not just that it motivates reconnection, but that it updates the parameters of the reward system itself.

1.2 Loneliness as a Regulatory Signal

Beyond its subjective character, loneliness appears to function as a regulatory signal — something that pushes behavior in particular directions in response to a perceived deficit. There is evidence that loneliness increases attention to social information, heightens sensitivity to interpersonal cues, and biases decision-making toward social reconnection (Cacioppo, Norris, Decety, Monteleone, & Nusbaum, 2009; Inagaki et al., 2016). These responses suggest a short-term adaptive function: notice the deficit, correct for it.

The more complicated case is what happens when the signal cannot be resolved. When discrepancies persist over time, the same mechanisms that promote reconnection may instead produce withdrawal, altered reward sensitivity, or compensatory shifts toward other reinforcement sources (Tomova et al., 2020). That transition — from short-term regulatory signal to longer-term learning-driven change — is where I think the prediction-error framing becomes most useful. A purely regulatory account does not say much about what happens when the regulation fails. An RL account provides at least a sketch of the mechanism.

1.3 Neural Encoding of Social Value

Whatever loneliness is doing computationally, its effects have to be implemented in neural systems. Converging evidence from social and affective neuroscience points to the ventral striatum and medial prefrontal cortex as key sites where social feedback — acceptance, rejection, peer evaluation — is encoded as value-relevant signal (Eisenberger, Lieberman, & Williams, 2003; Somerville, Heatherton, & Kelley, 2006; Powers, Somerville, Kelley, & Heatherton, 2013; Bhanji & Delgado, 2014). These are also the regions most consistently implicated in reward prediction error coding.

I want to be careful not to overstate how well-established the connection between social value encoding and standard reward circuitry is. There is ongoing debate about whether social outcomes engage the same neural machinery as nonsocial rewards, or whether they rely on partially distinct circuits (Bhanji & Delgado, 2014; Ruff & Fehr, 2014). The ASRRF assumes substantial overlap, which is a simplification that may need revision. What does seem reasonably clear is that these regions are involved in representing the motivational significance of social feedback and in updating expectations about social outcomes. That is enough to motivate the framework, even if the mechanistic details remain underspecified.

One way to summarize the claim: if loneliness reflects a sustained gap in perceived social value, its effects may not be limited to generating distress. They may involve progressive distortion of the brain's expectations about social reward — a shift in the running estimate of how much social interaction is likely to deliver. The pain of loneliness might be a symptom; the biased expectation might be the mechanism that actually shapes long-term behavior.

2 Adolescent Reward System Development

The reason I think adolescence matters for this account is not just that it is a sensitive period for social experience in some general sense — it is that the specific neural systems implicated in reward learning and prediction error processing appear to be undergoing significant change during this period. The literature on adolescent reward system development is large and I will not try to review it comprehensively here. What follows is a selective account of the features most relevant to the ASRRF.

2.1 Dopaminergic Sensitivity and Reward Plasticity

There is fairly consistent evidence that mesolimbic dopamine pathways are functionally different during adolescence than in adulthood. Adolescents show amplified neural responses to rewarding experiences in several paradigms, particularly in ventral striatum, compared with both children and adults (Casey, Jones, & Hare, 2008; Forbes & Dahl, 2012; Galván, 2013). This is often interpreted as reflecting ongoing calibration of valuation systems — the system is still settling, which makes it more responsive to input.

As illustrated in Figure 2, reward sensitivity appears to peak during mid-adolescence. I should note some caution here: the neuroimaging literature on adolescent reward sensitivity is not perfectly consistent, and the magnitude and direction of effects vary across paradigms and samples (Galván, 2013). What I take from this literature is a general principle rather than a precise claim: adolescent reward systems are in the process of calibration, and that process makes them more susceptible to being reshaped by experience than fully mature systems would be.

For the ASRRF, the implication is that repeated social mismatch during this period might produce disproportionate learning effects — not because adolescents are uniquely fragile, but because their valuation systems are still being calibrated and therefore more sensitive to input that drives updating.

2.2 Social Reward Salience in Adolescence

During adolescence, social feedback — particularly from peers — becomes an unusually potent source of reinforcement. Neuroimaging studies show heightened activation in reward-related regions including ventral striatum and medial prefrontal cortex in response to social approval and peer evaluation, compared with responses to nonsocial rewards (Davey, Allen, Harrison, Dwyer, & Yücel, 2010; Gunther Moor et al., 2010; Jones et al., 2014). This developmental shift in the motivational salience of social outcomes is well-documented, even if the exact mechanisms are not fully understood.

The relevance for loneliness is fairly direct: if social feedback has elevated motivational significance during adolescence, then disruptions to expected social reward may carry more weight during this period than they would in childhood or adulthood. The same objective social experience — being ignored, not getting a response, feeling excluded — may generate a stronger prediction error signal when social reward is more salient. That is at least a reasonable hypothesis, though I do not think the direct evidence for it is yet strong.

2.3 Reward-Guided Decision Making in Adolescence

There is also substantial evidence that adolescent decision-making is strongly guided by reward signals, particularly under uncertain or socially evaluative conditions (Steinberg, 2010; Van Duijvenvoorde et al., 2022). Adolescents tend to show increased willingness to explore uncertain options and to update choices based on reinforcement feedback. This enhanced reward-guided learning supports social exploration but also means that the behavioral consequences of recalibrated reward expectations could be broader than one might expect — extending beyond social decisions to shape motivational patterns more generally.

This is one of the more speculative aspects of the ASRRF. Whether reward recalibration driven by social prediction error actually generalizes to non-social behavioral patterns is an empirical question, not something I think can be resolved from the existing literature. I include it because it is a testable prediction, not because it is established.

3 Reward Prediction Error and Adaptive Learning

The concept of reward prediction error was, I think, the piece of this puzzle that took me longest to understand properly. I had encountered RPE before, but mostly as a neuroscience fact — dopamine neurons respond to unexpected rewards — without really thinking through what that means for learning over time. What I eventually understood, primarily from reading Schultz (2016) and Hauser et al. (2015), is that the prediction error signal is not just a response to good or bad outcomes. It is a teaching signal that drives systematic updating of the brain's predictions about future outcomes.

Positive prediction errors — outcomes that exceed expectations — strengthen the association between actions and rewards. Negative prediction errors — outcomes that fall short — downgrade expected value. Over repeated cycles, this updating process progressively brings predictions into alignment with the actual statistical structure of the environment. That is adaptive in most cases. The question the ASRRF is asking is what happens when the environment is persistently below expectation in a specific domain — social reward — and whether the updating that results is, in the long run, functional or dysfunctional.

3.1 Prediction Error as a Learning Signal

Reward prediction error functions as a learning signal by driving the brain's value representations toward better alignment with experience. Dopaminergic signaling within mesolimbic and frontostriatal circuits encodes discrepancies between expected and actual outcomes, allowing ongoing refinement of predictions about future reward (Hauser et al., 2015; Schultz, 2016). Through repeated cycles of prediction, feedback, and updating, reinforcement learning mechanisms shape motivational priorities and guide adaptive behavior.

One thing I find important to emphasize is that this process is not just about individual moments of disappointment or satisfaction. The updating is cumulative. Persistent patterns of prediction error — whether positive or negative — gradually recalibrate the expectations that the system brings to new situations. That cumulative recalibration is what makes chronic loneliness potentially different from episodic loneliness: not just that it feels worse, but that it may produce systematic changes in how the reward system represents social outcomes.

3.2 Developmental Modulation of Prediction Error

There is evidence that adolescents show amplified neural responses to feedback and outcome evaluation, consistent with increased weighting of prediction errors in learning (Hauser et al., 2015; Larsen & Luna, 2018). Most of this evidence comes from fMRI studies measuring BOLD responses rather than from formal computational model fits to behavior — a limitation I will return to in Section 6.6. The interpretation is that ongoing maturation of dopaminergic systems and frontostriatal connectivity alters how discrepancies between expectation and outcome are encoded and integrated.

If this interpretation is correct, it implies that adolescence may constitute a developmental window during which the effects of persistent prediction errors — including persistent social prediction errors — are disproportionately large. The same accumulated signal that produces modest updating in an adult might produce more substantial recalibration in an adolescent whose learning systems are still being calibrated. I think this is a plausible claim, but it rests on extrapolation rather than direct evidence. The direct evidence for developmentally varying sensitivity to social prediction error specifically is, to my knowledge, quite limited.

3.3 Social Prediction Error and Loneliness

Connecting social cognition to reinforcement learning raises the question of whether loneliness can be usefully modeled as a form of persistent social reward prediction error. Prior work has applied RL frameworks to social learning more broadly (Behrens, Hunt, Woolrich, & Rushworth, 2008; Hackel, Doll, & Amodio, 2015; Will, Rutledge, Moutoussis, & Dolan, 2017), establishing that social outcomes can drive prediction-error-based updating. The step I am proposing is to apply that logic specifically to the case of loneliness.

Loneliness arises when perceived social outcomes fall short of expectations. Within an RL framework, that shortfall constitutes a negative prediction error — a signal that expected value needs to be revised downward. If that signal is generated repeatedly, the learning system may progressively recalibrate expectations about social interaction: reducing anticipated reward, increasing uncertainty, or both. The behavioral consequence might be withdrawal — not as a choice, exactly, but as the output of a system that has updated its model of social reward.

I want to be careful about two things here. First, this account does not mean loneliness is nothing but a prediction error. The affective character of loneliness is real and matters. The claim is that the prediction-error framing captures an additional mechanistic dimension that affective accounts leave underspecified. Second, the direct evidence that chronic loneliness actually drives the kind of parametric RL changes I am describing is, to be honest, limited. The existing neuroimaging literature on loneliness and social reward is suggestive — Inagaki et al. (2016) found elevated ventral striatum activity to close others in lonely individuals, which is at least compatible with this account — but it does not directly measure RL parameters, and I think that gap matters.

4 The Adolescent Social Reward Recalibration Framework

4.1 Conceptual Overview

The ASRRF is a preliminary theoretical framework, not a finished theory. I want to be clear about that from the outset, partly because I think the loneliness literature sometimes suffers from

frameworks that are stated with more confidence than the evidence warrants. What I am proposing is a specific way of organizing the connections between loneliness, social prediction error, and adolescent reward plasticity — one that is explicit enough to generate testable predictions, but also tentative enough to be revised or abandoned if the evidence does not support it.

The core claim is that chronic loneliness during adolescence may function as a persistent pattern of negative sRPE that progressively recalibrates how social reward is anticipated and pursued. Because adolescent reward circuitry is still being calibrated, that recalibration process may be more consequential during this period than it would be in adulthood. The framework does not claim that loneliness is only a prediction error — it claims that the prediction-error component is mechanistically important and, crucially, measurable in ways that purely affective accounts are not.

One thing the ASRRF commits to that I think is important: recalibration, if it occurs, should be recoverable from behavior using formal RL modeling. That is a specific, potentially falsifiable claim. If a well-powered longitudinal study applied computational RL models to social feedback tasks in lonely and non-lonely adolescents and found no difference in learning-rate parameters, that would count against the framework. I think that kind of exposure to potential refutation is what this literature needs, and it is the main reason I find the prediction-error framing worth developing.

4.2 Mechanistic Pathway: From Social Discrepancy to Neural Recalibration

The proposed pathway, sketched as a sequence of steps, runs roughly as follows. Individuals form expectations about social belonging and interpersonal reward based on prior experience and developmental context. When social outcomes fall short of those expectations, negative social prediction errors are generated within reward-learning systems. Repeated exposure to such discrepancies engages dopaminergic mechanisms that update value representations associated with social interaction. Over time, if the discrepancies persist, those updates may accumulate into a more stable recalibration of incentive salience — altering how strongly social stimuli motivate approach behavior.

One question I kept returning to when developing this account was: why would the brain recalibrate downward rather than simply persist in trying to meet its social goals? The answer I find most plausible is that the learning system is doing what it is supposed to do — updating its model of the environment based on experience. If social interaction consistently fails to deliver expected reward, revising expectations downward is a locally rational response. The problem is that this locally rational updating can become self-reinforcing: reduced expectations lead to reduced engagement, which reduces the opportunities for positive social prediction errors that might reverse the recalibration.

Moderating variables. I expect the magnitude and direction of recalibration to vary substantially across individuals, and the framework identifies several candidate moderators. Baseline reward sensitivity — indexed by Behavioral Approach System scores or tonic dopaminergic markers — should influence how strongly the system responds to negative sRPE signals (Forbes et al., 2010; Nikolova et al., 2011). Rejection sensitivity (Downey & Feldman, 1996) may function as an inflated prior that amplifies the magnitude of negative prediction errors for any given objective social outcome. Pubertal timing matters because it affects when the

plasticity window opens relative to the development of prefrontal regulatory capacity (Forbes & Dahl, 2010). Dopaminergic genetic variation — DRD2, DRD4, COMT — may scale the steepness of value updates (Nikolova et al., 2011). Early adversity and attachment history shape the prior distribution over expected social outcomes. Sex-related differences in ventral striatal reactivity to peer feedback (Guyer et al., 2014) suggest differential recalibration trajectories may exist across individuals.

I include this moderator list not because I think it is complete, but because it illustrates that the framework predicts substantial between-person variability — and that this variability should have a specific, theoretically grounded structure. Whether these moderators actually account for variability in the ways predicted is an empirical question.

4.3 Divergent Developmental Pathways

One of the more speculative predictions of the ASRRF is that reward recalibration may produce divergent developmental trajectories depending on environmental context and individual learning history. One possible pathway involves progressive reduction in expected social reward — leading toward diminished motivation for social engagement, withdrawal, and blunted reward responsiveness. Another possible pathway involves compensatory upregulation of sensitivity to nonsocial or high-intensity rewards — leading toward risk-taking, novelty-seeking, or other alternative reinforcement strategies.

I want to be explicit that this divergence is speculative. I am reasoning from the general logic of RL models about what should happen when one reward domain is chronically underdelivering, but I do not think the empirical literature on lonely adolescents has directly tested whether these two pathways exist as distinct trajectories, or whether the moderators that predict one pathway versus another are what the framework would suggest. This is an area where future empirical work is genuinely needed before any strong claims can be made.

4.4 Integration with Developmental Decision-Making

If social reward recalibration occurs, it may not be contained to decisions about social interaction. Because adolescent decision-making relies heavily on reinforcement-based learning (Steinberg, 2010; Van Duijvenvoorde et al., 2022), shifts in expected social reward could plausibly bias cost-benefit evaluation more broadly — increasing sensitivity to rejection costs, decreasing the perceived value of social engagement, and potentially generalizing to non-social contexts where social comparison is implicated.

How far this generalization extends is something I am uncertain about. The claim that reward recalibration affects broad motivational patterns, rather than just social behavior specifically, is an extrapolation from the general logic of RL models rather than a finding from the loneliness literature. It is a testable prediction, and I think it is worth testing, but I would not want it to be read as established.

4.5 Testable Predictions and Theoretical Implications

The ASRRF generates four predictions that I think are empirically separable from those of alternative accounts. I list them in rough order of the confidence I have in them, not in order of how easy they would be to test.

Prediction 1. Chronic loneliness should be associated with attenuated ventral striatal response to anticipated rather than received social reward. This prediction distinguishes the present account from accounts that predict heightened threat sensitivity or altered responses to negative outcomes specifically. The evidence I find most suggestive here comes from studies showing that lonely individuals show altered ventral striatal reactivity to social stimuli (Inagaki et al., 2016), though those studies do not directly test the anticipation versus receipt distinction.

Prediction 2. Longitudinal exposure to social prediction error should predict changes in formally estimated RL parameters — particularly learning rate α and inverse temperature β — in social feedback tasks. This is the prediction I think is most important to the framework and also the most technically demanding to test, since it requires formal computational modeling of choice behavior rather than just neuroimaging.

Prediction 3. Interventions that modify perceived social expectations — by making social feedback more predictable and positive — should influence reward learning dynamics and produce measurable changes in RL parameters. If this prediction fails, it would suggest either that the framework is wrong or that the intervention window was missed.

Prediction 4. All three effects above should be larger in mid-adolescence than in adulthood. This is the developmentally specific prediction that most clearly distinguishes the ASRRF from lifespan-invariant accounts. It is also, I think, the prediction whose failure would most cleanly argue against the framework as stated. If the effects are equally present in adults, the adolescent-specific plasticity argument loses much of its force.

I should be honest that none of these predictions has been tested directly. The framework is proposed in the context of indirect and partially consistent evidence, not as a post-hoc summary of established findings. Prediction 1 has the most existing support. Predictions 3 and 4 are the most genuinely speculative.

5 Implications for Behavior and Development

5.1 Motivational Adaptation and Social Engagement

If the ASRRF is approximately correct, some patterns of behavior commonly observed in lonely adolescents may be interpretable as learning-driven adjustments rather than social deficits. Reduced motivation to initiate social engagement, diminished responsiveness to social feedback, and withdrawal from peer activities could reflect a reward system that has updated its expectations about social outcomes — not a system that is broken or incapable of social behavior.

That reinterpretation has some clinical relevance. It suggests that the relevant intervention target might not be motivation per se, but the conditions under which motivation is generated. If the learning system has recalibrated expectations downward, then providing social opportunities without changing the expectation structure may not produce much behavioral change. But that is speculative — this kind of claim requires testing before it should influence intervention design.

5.2 Decision-Making and Risk Sensitivity

Changes in reward valuation may extend beyond social contexts to influence broader patterns of adolescent decision-making. Adolescents experiencing persistent loneliness may become more sensitive to rejection costs or less responsive to anticipated social benefits. Some may shift toward compensatory engagement with nonsocial high-intensity rewards, consistent with the divergent pathways described in Section 4.3 (Steinberg, 2010). These patterns are at least consistent with developmental findings on reward processing and adolescent decision strategies, though whether they specifically reflect the RL mechanism the framework proposes remains to be tested.

5.3 Identity Formation and Developmental Trajectories

Adolescence is also a period of significant identity development, during which social feedback plays an important role in shaping self-concept and interpersonal expectations. If social prediction error progressively recalibrates how social reward is anticipated, that recalibration may eventually extend to stable beliefs about belonging and social value — not just moment-to-moment motivation, but something closer to a schema about what social interaction is likely to deliver.

I think this is one of the more important longer-term implications of the framework, even though it is also one of the hardest to operationalize. The shift from trial-by-trial updating to stable prior is not something that happens at a single identifiable moment — it is the accumulated output of a learning process. Longitudinal studies that track both RL parameters and self-concept measures over adolescence would be the most direct way to examine whether this developmental drift actually occurs.

5.4 Implications for Mental Health Vulnerability

One reason I think the prediction-error framing of loneliness matters clinically is that it offers a somewhat specific account of why chronic loneliness and adolescent depression so often co-occur (Forbes & Dahl, 2012; Cacioppo, Hawkley, & Thisted, 2010). Anhedonia and reduced motivation — core features of depression — could, on this account, arise partly from downward recalibration of reward expectations across domains, not just social ones. The RL machinery that processes social prediction error is not socially specific; recalibration in that domain might generalize.

I want to be careful not to overclaim here. The relationship between loneliness and depression is well-established empirically, but the mechanism I am proposing is speculative. And even if the RL account were correct, loneliness is clearly not the only pathway to adolescent depression. What the framework suggests is one specific mechanism — recalibrated social reward expectations during a plasticity-sensitive period — that might be worth targeting in prevention efforts. But that suggestion requires empirical support before it should drive clinical practice.

6 Future Directions

6.1 Longitudinal Investigation of Social Prediction Error

The most direct test of the ASRRF would require longitudinal designs that track both behavioral measures of social experience and computational RL parameters over the course of adolescence. Cross-sectional comparisons of lonely and non-lonely adolescents can establish association but cannot establish whether recalibration is progressive, as the framework predicts. Repeated-measures designs that estimate individual RL parameters at multiple time points — fitting models to choice behavior in social feedback tasks — would be most informative. This is technically demanding but feasible with existing tools.

6.2 Computational Modeling of Social Learning

A related priority is the development of validated social RL tasks that can be used to estimate learning-rate parameters and other RL quantities from behavior, rather than relying primarily on neuroimaging proxies. Most existing work on adolescent prediction error processing has inferred RPE signals from BOLD responses; it is not clear that BOLD changes and formal RL parameters are measuring the same thing. Studies that collect both behavioral choice data and neuroimaging data, and that estimate RL parameters from the former, would help clarify how the different levels of analysis relate (Behrens et al., 2008; Will et al., 2017).

6.3 Experimental Manipulation of Social Expectations

Experimental paradigms that systematically manipulate perceived social expectations — through controlled feedback about social outcomes, simulated peer interaction, or other methods — would allow more direct testing of the causal claims in the ASRRF. If the framework is correct, experimentally inducing sustained negative social prediction errors should produce measurable changes in anticipatory reward signals and in RL parameter estimates. Demonstrating that the direction of recalibration can be experimentally controlled would be strong evidence for the mechanistic account.

6.4 Intervention and Prevention Implications

If the ASRRF's core mechanism is supported empirically, it would suggest some specific priorities for intervention. Programs that target emotional symptoms of loneliness without addressing the learning environment that generates persistent social prediction errors may be addressing the symptom rather than the mechanism. An intervention designed around the ASRRF might focus on increasing the predictability and consistency of positive social feedback during adolescence — not just increasing social exposure, but changing the statistical structure of social outcomes in ways that allow the recalibration to run in the corrective direction.

I should be clear that this is theoretical reasoning from the framework, not a tested intervention strategy. The distance between a computational framework and an effective clinical program is substantial, and I do not want to overstate how much the ASRRF implies about what should actually be done.

6.5 Integrating Social, Developmental, and Neural Levels

One of the harder empirical questions is whether the recalibration effects the ASRRF describes actually occur in response to real-world social environments, or only in laboratory paradigms with controlled feedback. Laboratory tasks allow precise manipulation of prediction error signals, but they may not capture the complexity and variability of naturally occurring social interaction. Ecological momentary assessment approaches, combined with periodic laboratory RL tasks and neuroimaging, might help bridge that gap. If real-world social experience does not produce the parameter changes observed in the lab, the framework may be limited to describing laboratory phenomena — which would significantly constrain its broader implications.

6.6 Limitations and Boundary Conditions

I want to be explicit about several limitations that I think are important and that constrain how confidently the framework should be stated.

First, most developmental evidence for prediction-error processing in adolescents comes from BOLD responses rather than formal model fits to choice behavior. Whether BOLD changes in ventral striatum correspond to the RL quantities the framework assumes is not established — it is an inference. Future work should prioritize behavioral data that can be directly modeled.

Second, the framework assumes that loneliness produces a sustained negative sRPE rather than a transient deflection. Demonstrating that the signal is sustained — rather than reflecting a stable trait difference in social reward sensitivity that predates loneliness — requires longitudinal within-person modeling that, to my knowledge, has not been done.

Third, the assumption that social and nonsocial reward share core dopaminergic machinery is a simplification. The evidence for shared versus distinct processing of social and nonsocial reward is mixed (Bhanji & Delgado, 2014; Ruff & Fehr, 2014). If social reward processing is substantially distinct, the RL framework may need significant modification to apply.

Fourth, the framework does not currently specify the boundary between adaptive recalibration and pathological recalibration. The same mechanism that produces appropriate updating in a context of genuinely limited social reward might produce maladaptive updating in a context where social reward is available but anticipated poorly. Identifying what determines which outcome occurs — and at what point updating becomes pathological — is important for both theoretical completeness and clinical application.

Fifth, and more broadly: the ASRRF is a theoretical framework built from indirect evidence. Its predictions are plausible given what we know about adolescent reward systems and loneliness, but none of them has been directly tested. I think the framework is worth developing and testing, but I would not want it to be read as a summary of established findings.

Conclusion

I started thinking about this topic because I was frustrated by how abstract most explanations of loneliness felt. Saying that loneliness is a social alarm, or that it arises from unmet belonging needs, tells you something — but it does not tell you much about what the brain is actually doing differently in a person who has been chronically lonely for two years versus someone who has

not. The prediction-error framing felt like it might offer something more mechanistically specific.

What I have tried to do in this paper is develop that intuition into a preliminary framework with enough structure to generate testable predictions. The ASRRF proposes that chronic loneliness during adolescence can be modeled as a persistent pattern of negative social reward prediction error that recalibrates how social outcomes are anticipated — and that this recalibration is more consequential during adolescence, when dopaminergic systems are still being calibrated, than it would be in adulthood. The framework makes four predictions that are, in principle, empirically separable from those of existing accounts.

I am aware that a lot of what I have argued here is extrapolation and theoretical inference rather than direct evidence. The existing literature supports pieces of the account — the developmental profile of adolescent reward sensitivity, the role of social feedback in shaping ventral striatal activity, the general applicability of RL models to social learning — but it does not directly test the specific causal chain the ASRRF proposes. I think that is honest, and I think it is the right reason to develop the framework: not because the evidence already supports it, but because it generates predictions that could change how we understand loneliness if they held up.

Whether they do is something I hope to see tested. The hardest version of that test would be a longitudinal study that formally estimates RL parameters from social feedback behavior at multiple points across adolescence, in participants with varying levels of chronic loneliness, with the developmentally timed comparison built in. If that study found nothing — no difference in learning-rate parameters, no adolescent-specific effect — I would take that seriously as evidence against the framework. That kind of falsifiability seems worth aiming for.

References

- Behrens, T. E. J., Hunt, L. T., Woolrich, M. W., & Rushworth, M. F. S. (2008). Associative learning of social value. *Nature*, 456(7219), 245–249. <https://doi.org/10.1038/nature07538>
- Bhanji, J. P., & Delgado, M. R. (2014). The social brain and reward: Social information processing in the human striatum. *Wiley Interdisciplinary Reviews: Cognitive Science*, 5(1), 61–73. <https://doi.org/10.1002/wcs.1266>
- Cacioppo, J. T., & Hawkley, L. C. (2009). Perceived social isolation and cognition. *Trends in Cognitive Sciences*, 13(10), 447–454. <https://doi.org/10.1016/j.tics.2009.06.005>
- Cacioppo, J. T., Cacioppo, S., & Boomsma, D. I. (2014). Evolutionary mechanisms for loneliness. *Cognition and Emotion*, 28(1), 3–21. <https://doi.org/10.1080/02699931.2013.837379>
- Cacioppo, J. T., Hawkley, L. C., & Thisted, R. A. (2010). Perceived social isolation makes me sad: Five year cross-lagged analyses of loneliness and depressive symptomatology. *Psychology and Aging*, 25(2), 453–463. <https://doi.org/10.1037/a0017216>
- Cacioppo, J. T., Norris, C. J., Decety, J., Monteleone, G., & Nusbaum, H. (2009). In the eye of the beholder: Individual differences in perceived social isolation predict regional brain activation to social stimuli. *Journal of Cognitive Neuroscience*, 21(1), 83–92. <https://doi.org/10.1162/jocn.2009.21007>

Casey, B. J., Jones, R. M., & Hare, T. A. (2008). The adolescent brain. *Annals of the New York Academy of Sciences*, 1124(1), 111–126. <https://doi.org/10.1196/annals.1440.010>

Davey, C. G., Allen, N. B., Harrison, B. J., Dwyer, D. B., & Yücel, M. (2010). Being liked activates primary reward and midline self-related brain regions. *Human Brain Mapping*, 31(4), 660–668. <https://doi.org/10.1002/hbm.20895>

Downey, G., & Feldman, S. I. (1996). Implications of rejection sensitivity for intimate relationships. *Journal of Personality and Social Psychology*, 70(6), 1327–1343. <https://doi.org/10.1037/0022-3514.70.6.1327>

Eisenberger, N. I., Lieberman, M. D., & Williams, K. D. (2003). Does rejection hurt? An fMRI study of social exclusion. *Science*, 302(5643), 290–292. <https://doi.org/10.1126/science.1089134>

Forbes, E. E., & Dahl, R. E. (2010). Pubertal development and behavior: Hormonal activation of social and motivational tendencies. *Brain and Cognition*, 72(1), 66–72. <https://doi.org/10.1016/j.bandc.2009.10.012>

Forbes, E. E., & Dahl, R. E. (2012). Research review: Altered reward function in adolescent depression: What, when and how? *Journal of Child Psychology and Psychiatry*, 53(1), 3–15. <https://doi.org/10.1111/j.1469-7610.2011.02477.x>

Forbes, E. E., Ryan, N. D., Phillips, M. L., Manuck, S. B., Worthman, C. M., Moyles, D. L., Tarr, J. A., Sciarillo, S. R., & Dahl, R. E. (2010). Healthy adolescents' neural response to reward: Associations with puberty, positive affect, and depressive symptoms. *Journal of the American Academy of Child & Adolescent Psychiatry*, 49(2), 162–172.

Galván, A. (2010). Adolescent development of the reward system. *Frontiers in Human Neuroscience*, 4, 6. <https://doi.org/10.3389/neuro.09.006.2010>

Galván, A. (2013). The teenage brain: Sensitivity to rewards. *Current Directions in Psychological Science*, 22(2), 88–93. <https://doi.org/10.1177/0963721413480859>

Gunther Moor, B., van Leijenhorst, L., Rombouts, S. A. R. B., Crone, E. A., & Van der Molen, M. W. (2010). Do you like me? Neural correlates of social evaluation and developmental trajectories. *Social Neuroscience*, 5(5–6), 461–482. <https://doi.org/10.1080/17470910903526155>

Guyer, A. E., Caouette, J. D., Lee, C. C., & Ruiz, S. K. (2014). Will they like me? Adolescents' emotional responses to peer evaluation. *International Journal of Behavioral Development*, 38(2), 155–163. <https://doi.org/10.1177/0165025413515627>

Guyer, A. E., Silk, J. S., & Nelson, E. E. (2016). The neurobiology of the emotional adolescent: From the inside out. *Neuroscience & Biobehavioral Reviews*, 70, 74–85. <https://doi.org/10.1016/j.neubiorev.2016.07.014>

Hackel, L. M., Doll, B. B., & Amodio, D. M. (2015). Instrumental learning of traits versus rewards: Dissociable neural correlates and effects on choice. *Nature Neuroscience*, 18(9), 1233–1235. <https://doi.org/10.1038/nn.4080>

Hauser, T. U., Iannaccone, R., Walitza, S., Brandeis, D., & Brem, S. (2015). Cognitive flexibility in adolescence: Neural and behavioral mechanisms of reward prediction error processing in

adaptive decision making during development. *NeuroImage*, 104, 347–354.
<https://doi.org/10.1016/j.neuroimage.2014.09.018>

Inagaki, T. K., Muscatell, K. A., Moieni, M., Dutcher, J. M., Jevtic, I., Irwin, M. R., & Eisenberger, N. I. (2016). Yearning for connection? Loneliness is associated with increased ventral striatum activity to close others. *Social Cognitive and Affective Neuroscience*, 11(7), 1096–1101. <https://doi.org/10.1093/scan/nsv076>

Jones, R. M., Somerville, L. H., Li, J., Ruberry, E. J., Powers, A., Mehta, N., Dyke, J., & Casey, B. J. (2014). Adolescent-specific patterns of behavior and neural activity during social reinforcement learning. *Cognitive, Affective, & Behavioral Neuroscience*, 14(2), 683–697.
<https://doi.org/10.3758/s13415-014-0257-z>

Larsen, B., & Luna, B. (2018). Adolescence as a neurobiological critical period for the development of higher-order cognition. *Neuroscience & Biobehavioral Reviews*, 94, 179–195.
<https://doi.org/10.1016/j.neubiorev.2018.09.005>

Masten, C. L., Eisenberger, N. I., Borofsky, L. A., Pfeifer, J. H., McNealy, K., Mazziotta, J. C., & Dapretto, M. (2009). Neural correlates of social exclusion during adolescence: Understanding the distress of peer rejection. *Social Cognitive and Affective Neuroscience*, 4(2), 143–157.
<https://doi.org/10.1093/scan/nsp007>

Nikolova, Y. S., Ferrell, R. E., Manuck, S. B., & Hariri, A. R. (2011). Multilocus genetic profile for dopamine signaling predicts ventral striatum reactivity. *Neuropsychopharmacology*, 36(9), 1940–1947. <https://doi.org/10.1038/npp.2011.82>

Powers, K. E., Somerville, L. H., Kelley, W. M., & Heatherton, T. F. (2013). Rejection sensitivity polarizes striatal-medial prefrontal activity when anticipating social feedback. *Journal of Cognitive Neuroscience*, 25(11), 1887–1895. https://doi.org/10.1162/jocn_a_00446

Ruff, C. C., & Fehr, E. (2014). The neurobiology of rewards and values in social decision making. *Nature Reviews Neuroscience*, 15(8), 549–562. <https://doi.org/10.1038/nrn3776>

Schreuders, E., Braams, B. R., Blankenstein, N. E., Peper, J. S., Güroğlu, B., & Crone, E. A. (2018). Contributions of reward sensitivity to ventral striatum activity across adolescence and early adulthood. *Child Development*, 89(3), 797–810. <https://doi.org/10.1111/cdev.13056>

Schultz, W. (2016). Dopamine reward prediction error coding. *Dialogues in Clinical Neuroscience*, 18(1), 23–32.

Somerville, L. H., Heatherton, T. F., & Kelley, W. M. (2006). Anterior cingulate cortex responds differentially to expectancy violation and social rejection. *Nature Neuroscience*, 9(8), 1007–1008. <https://doi.org/10.1038/nn1728>

Steinberg, L. (2010). A dual systems model of adolescent risk-taking. *Developmental Psychobiology*, 52(3), 216–224. <https://doi.org/10.1002/dev.20445>

Tomova, L., Wang, K. L., Thompson, T., Matthews, G. A., Takahashi, A., Tye, K. M., & Saxe, R. (2020). Acute social isolation evokes midbrain craving responses similar to hunger. *Nature Neuroscience*, 23(12), 1597–1605. <https://doi.org/10.1038/s41593-020-00742-z>

Van Duijvenvoorde, A. C. K., van Hoorn, J., & Blankenstein, N. E. (2022). Risks and rewards in adolescent decision-making. *Current Opinion in Psychology*, 48, 101457.
<https://doi.org/10.1016/j.copsyc.2022.101457>

Will, G.-J., Rutledge, R. B., Moutoussis, M., & Dolan, R. J. (2017). Neural and computational processes underlying dynamic changes in self-esteem. *eLife*, 6, e28098.
<https://doi.org/10.7554/eLife.28098>